## GEAR SHIFT INDICATOR [SKYACTIV-D 1.5]

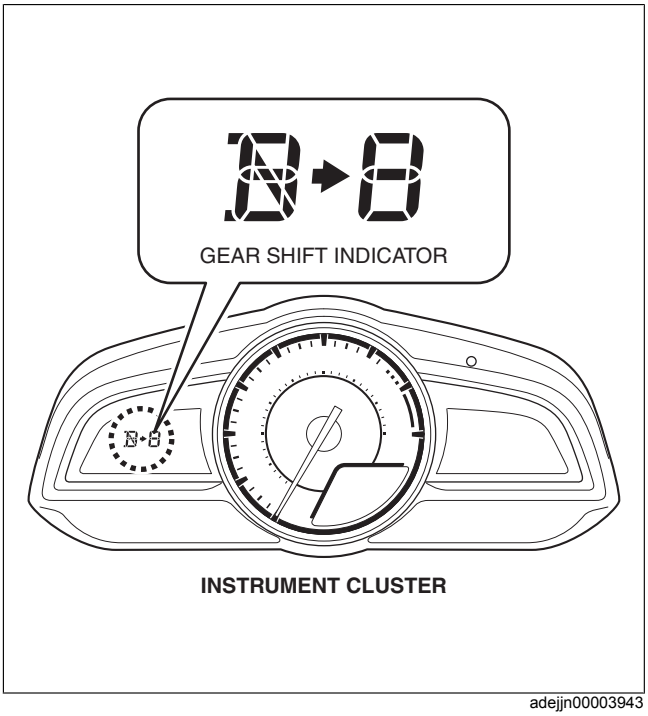
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**Purpose/Function**

- The gear shift indicator, built into the instrument cluster, informs the driver that a gear position or shift up/down operation is recommended according to the driving conditions.
- The PCM sends a gear shift indicator illumination request to the instrument cluster via CAN communication.

**Construction**

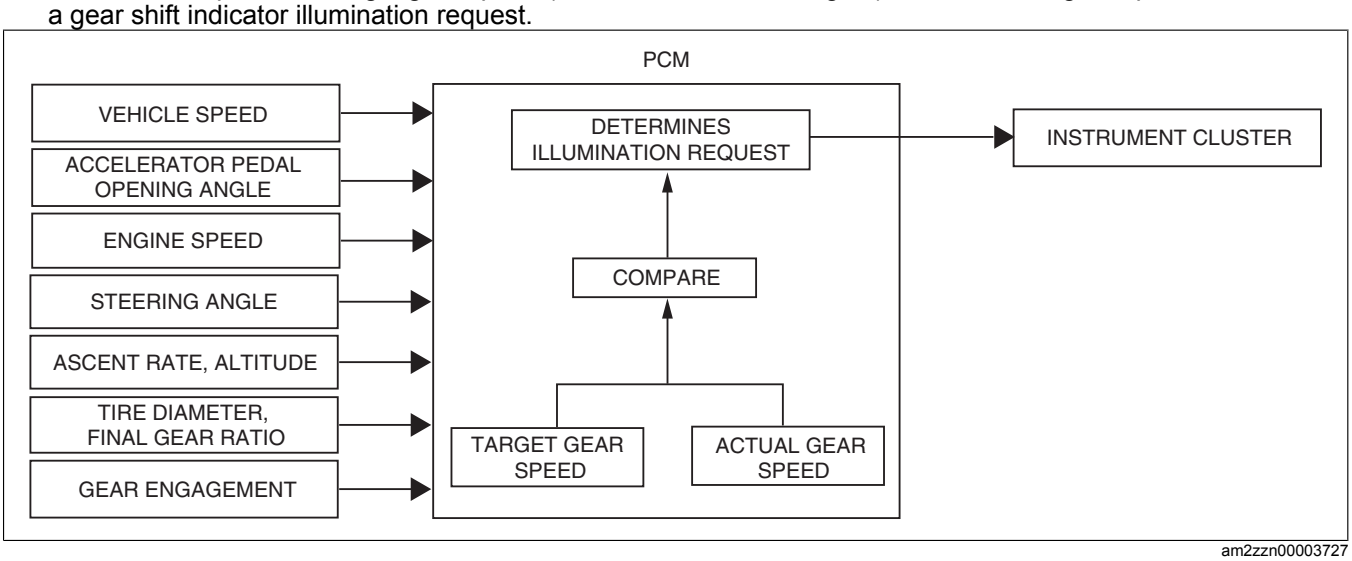
- The gear shift indicator is built into the instrument cluster.




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**Operation**

- The PCM compares the target gear speed (calculated from each signal) and the actual gear speed to determine a gear shift indicator illumination request.




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**Target gear speed (recommended gear position)**

— The target gear speed is determined by the vehicle speed, accelerator pedal opening angle, ascent rate, altitude, and steering angle.

**Actual gear speed (current gear position)**

The actual gear speed is determined by the vehicle speed, engine speed, tire diameter, final gear ratio, and gear engagement.

- The gear shift indicator illumination/off conditions are as follows:

**Gear shift indicator illumination condition**

- There are three gear shift indicator illumination patterns; normal operation, shift-down recommended operation, and shift-up recommended operation.

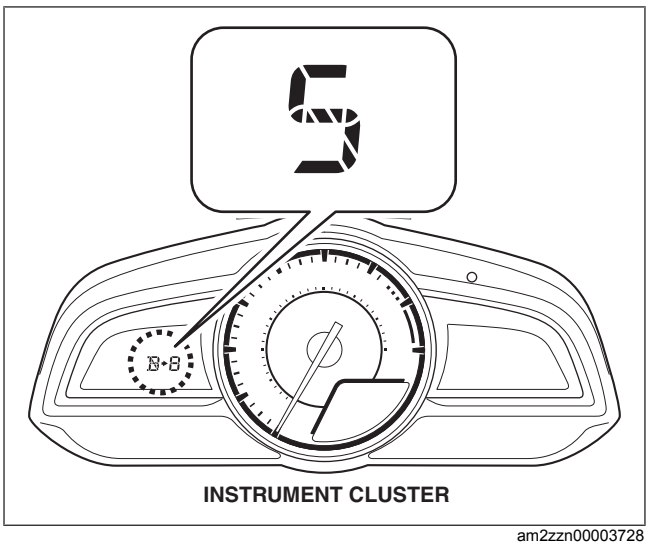
  **Normal operation**

  — When the following condition is met, the currently selected gear position is indicated.
    - Target gear speed and actual gear speed are equal

**Example: the following figure indicates that the currently selected gear position is 5th gear.**

**Shift-down recommended operation**

— If a shift-down operation is recommended when the following condition is met (determines smooth driveability is difficult), the recommended gear position is indicated at the same time.
  - Target gear speed is lower than actual gear speed

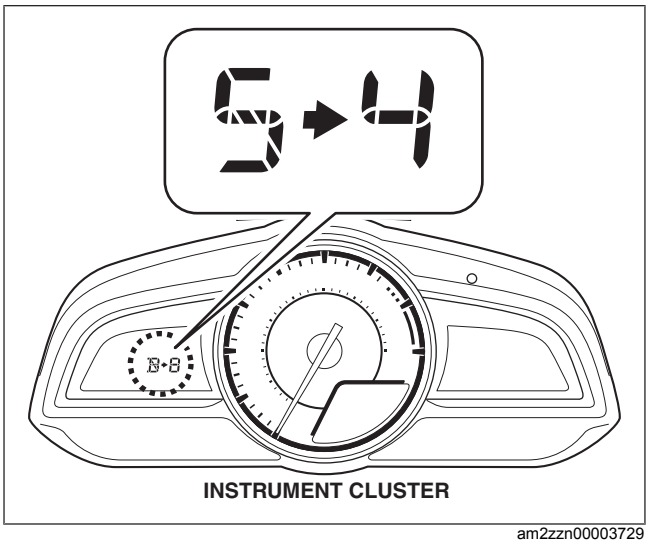



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**Example: The following figure indicates that a shift-down operation to 4th gear is recommended.**

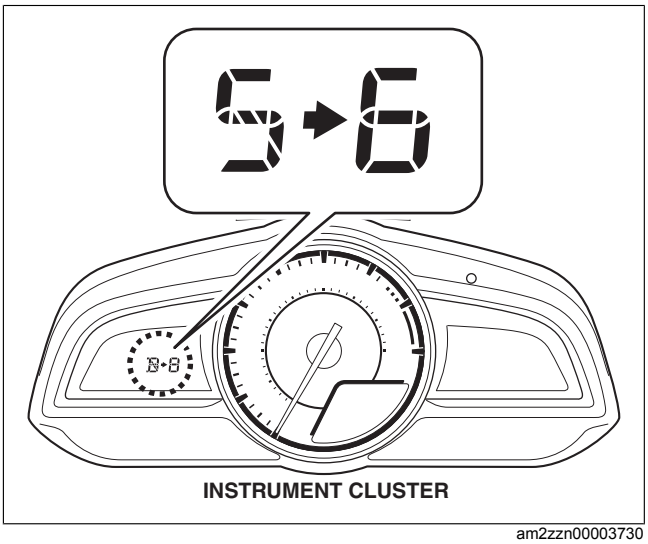
**Shift-up recommended operation**

— If a shift-up operation is recommended when the following condition is met (determines fuel economy worsens), the recommended gear position is indicated at the same time.
  - Target gear speed is higher than actual gear speed




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**Example: The following figure indicates that a shift-up operation to 6th gear is recommended.**




INSTRUMENT CLUSTER

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**Gear shift indicator off condition**

- When any one of the following conditions is met, the gear shift indicator turns off.
  - — Vehicle is stopped
  - — Gear is in neutral position or reverse position
  - — Until clutch is completely engaged when vehicle is started
  - — Clutch pedal is depressed for **2 s or more** while driven
- If the vehicle is driven with the gear not changed when a shift-down/up operation is recommended, the gear shift indicator may temporarily turn off.

**Fail-safe**

- Not applicable